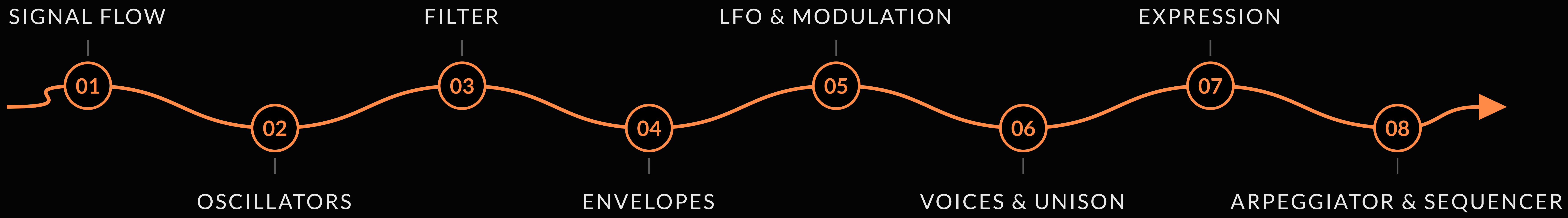
INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm



SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

TODAY

Club Control, Bucharest

SINTEZAUR

Bucharest's community for people who love synthesis. Your guide today: Iulian.

ZEEDO

Our retail partner, and the reason there are a dozen synths in this room.

SEQUENTIAL FOURM

Today's instrument: a 4-voice analog synth that feels pressure in every key.

SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

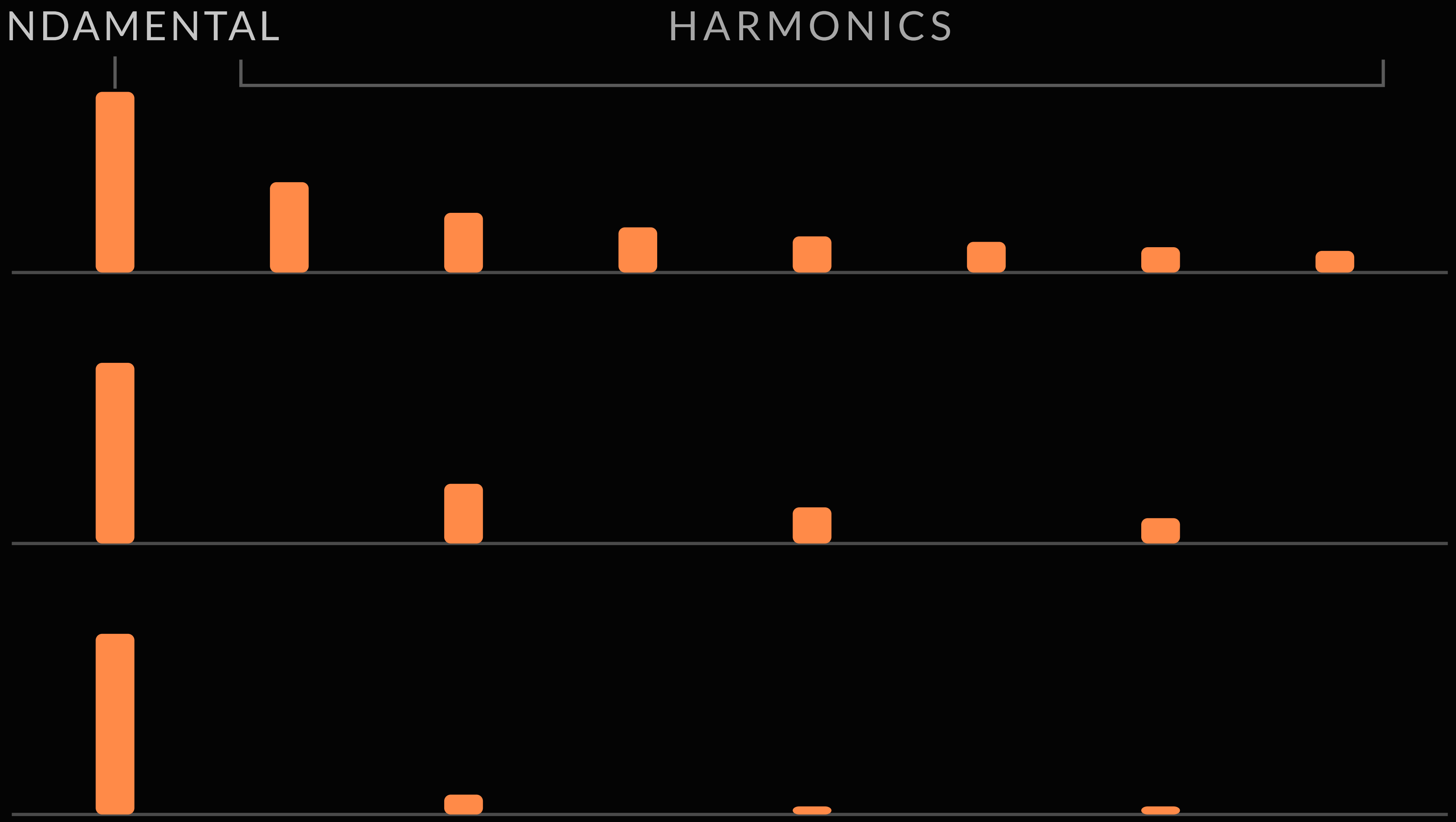
SIGNAL FLOW

How sound travels through a subtractive synthesizer

WHAT'S IN A SOUND

A note is a fundamental plus a stack of quieter harmonics. Their mix is the timbre.

BAR HEIGHT = HOW LOUD · LEFT TO RIGHT = HIGHER IN PITCH



SAW
EVERY HARMONIC
bright, buzzy

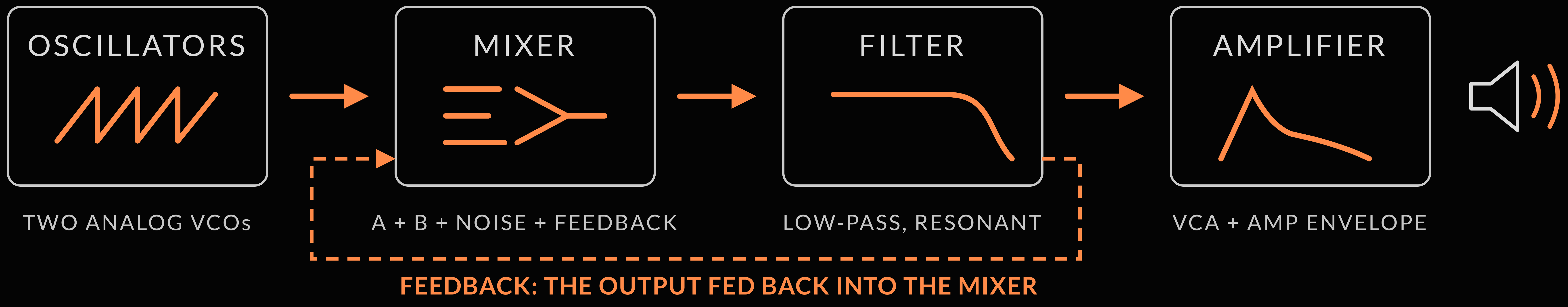
SQUARE
ODD HARMONICS ONLY
hollow, woody

TRIANGLE
ODD, FADING FAST
soft, nearly a sine

SQUARE IS PULSE AT 50%. OFF 50%, EVEN HARMONICS APPEAR — THAT'S WHAT PWM MOVES.

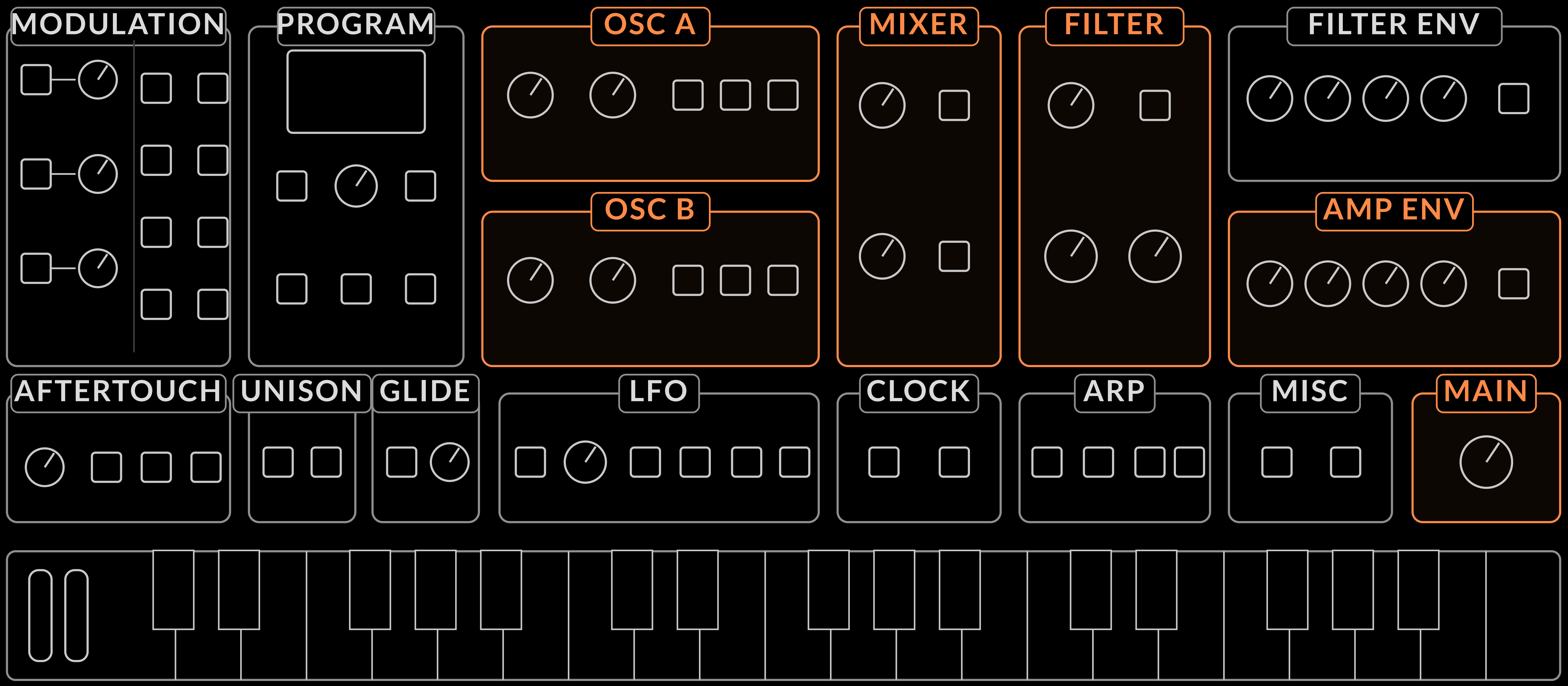
THE VOICE PATH

Oscillators, mixer, filter, amplifier: one road for every sound



WHERE ON FOURM

The signal path, left to right across the panel



THE MIXER

Every voice starts here: four sources feed the filter

OSC A

SAW + PULSE
THE BRIGHT BACKBONE

OSC B

TRI + SAW + PULSE
WEIGHT, DETUNE, LO MODES

NOISE

WHITE / PINK / VIOLET
AIR, PERCUSSION, TEXTURE

FEEDBACK

OUTPUT LOOP
SOFT-CLIP SATURATION

TURN UP AT LEAST ONE SOURCE, OR LET RESONANCE SELF-OSCILLATE

KEY TERMS

New words from this module

SIGNAL FLOW
The road a sound takes: oscillators → mixer → filter → amplifier → output.

VCO · OSCILLATOR
Voltage-controlled oscillator. Generates the raw waveform. Fourm has two.

MIXER
Sets the level of each source: Osc A, Osc B, noise, feedback.

HARMONICS
Quieter overtones above the fundamental; their mix is timbre.

NOISE
Unpitched sound, in white, pink, or violet flavors.

FEEDBACK
The output fed back into the mixer, soft-clipped: warmth, then growl.

VCF · VCA
Voltage-controlled filter and amplifier: the tone and loudness stages.

OSCILLATORS

Where every sound is born

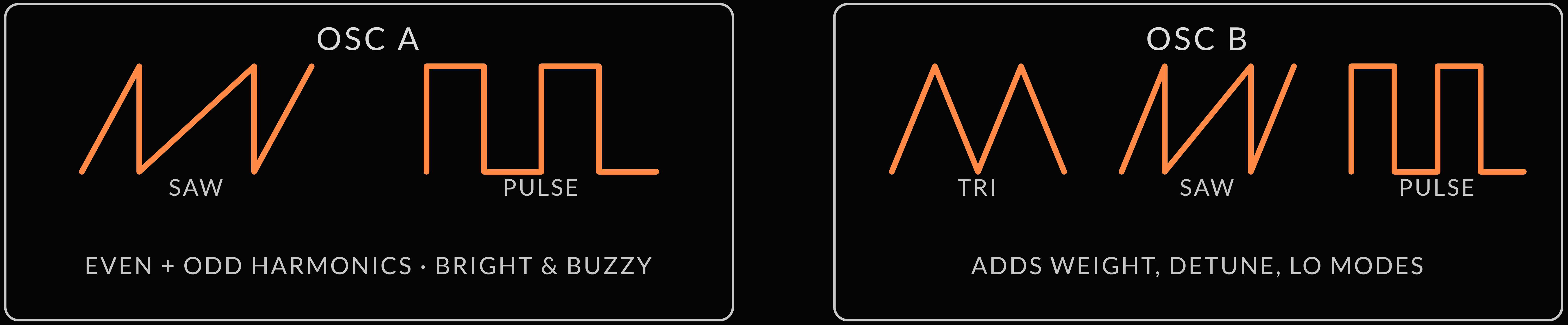
SAW

PULSE

TRI

THE WAVESHAPES

Two oscillators, five buttons, endless timbres



PRESS SEVERAL SHAPE BUTTONS AT ONCE: THE WAVES COMBINE

OCTAVES & TUNING

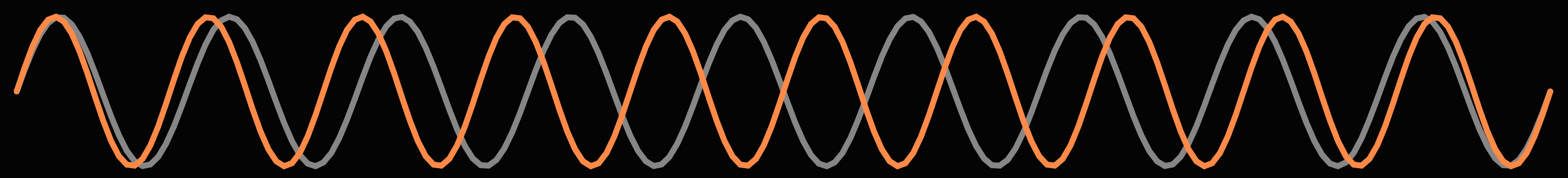
Detune B against A and the sound starts to breathe.

-2 | -1 | 0 | +1 | +2

FIVE OCTAVE POSITIONS ON EACH OSCILLATOR

OSC B ONLY: LO1 · LO2, BELOW THE OCTAVES —
SUB-AUDIO, IT BECOMES A MODULATOR (LEVEL 2)

FREQUENCY · FINE TUNE · ±7 SEMITONES



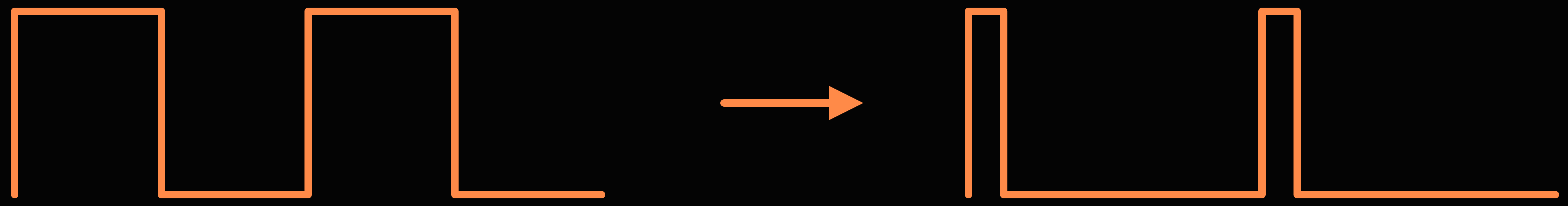
A AND B, A FEW CENTS APART

TOGETHER THEY BEAT — THE SOUND BREATHES

TOO FAR AND YOU HEAR TWO NOTES — BRING IT BACK UNTIL IT JUST SHIMMERS

PULSE WIDTH & PWM

No PW knob: hold PULSE, turn SELECT. 0 = square. Toward ±100: thin, then silence.



PW 0: SQUARE (50%)

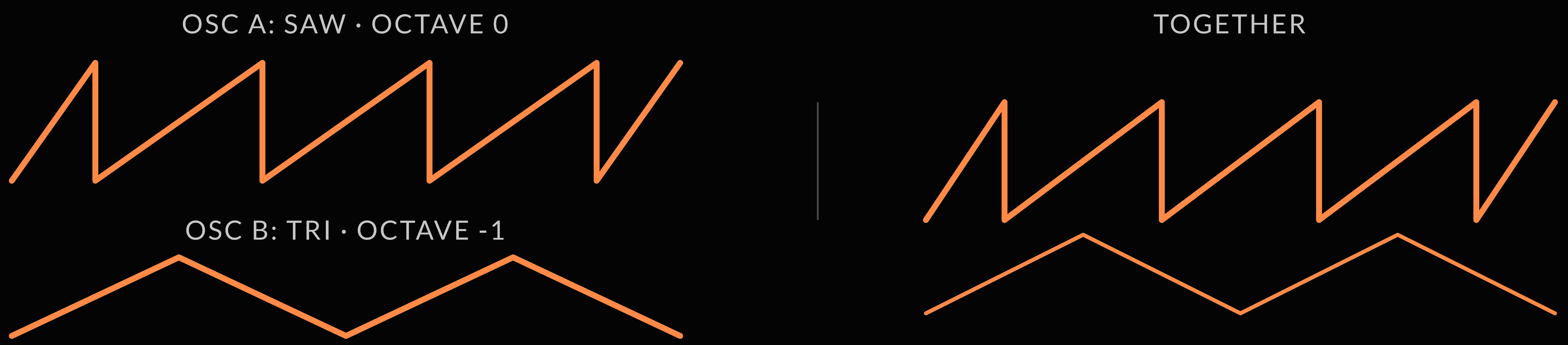
PW NEAR ±100%: THIN, REEDY

AT FULL ±100% THE PULSE NARROWS TO SILENCE

ROUTE LFO OR FILT ENV TO PW: THAT MOVEMENT IS PWM

THE SUB TRICK

Osc B, one octave down, triangle

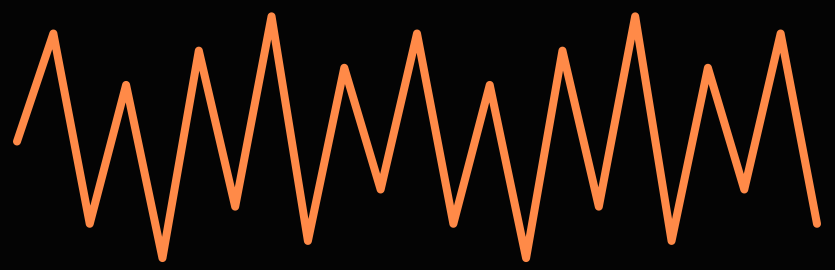


THE TRIANGLE REINFORCES THE FUNDAMENTAL: WEIGHT WITHOUT MUD

NOISE

Three colors of texture: white, pink, violet

WHITE

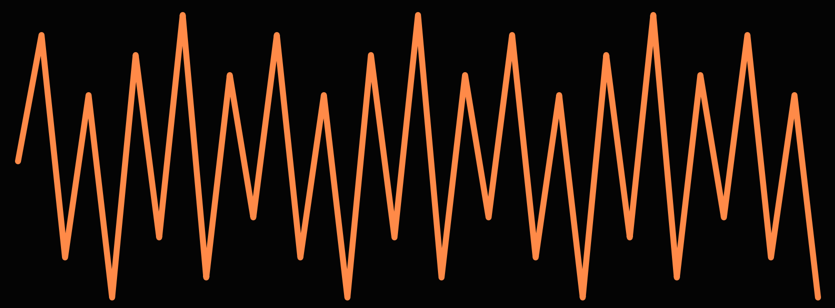


FLAT: HISS, SNARE

PINK

WARM: WIND, WAVES

VIOLET



BRIGHT: AIR, SIZZLE

HOLD NOISE + TURN SELECT FOR LEVEL · TYPE LIVES IN THE PARAM MENU

KEY TERMS

New words from this module

WAVESHAPE
The geometric shape of one cycle: it sets the raw timbre.

PULSE WIDTH (PW)
Duty cycle of the pulse wave, -100% to +100%. 0 = square.

PWM
Pulse width modulation: moving PW with the LFO or an envelope.

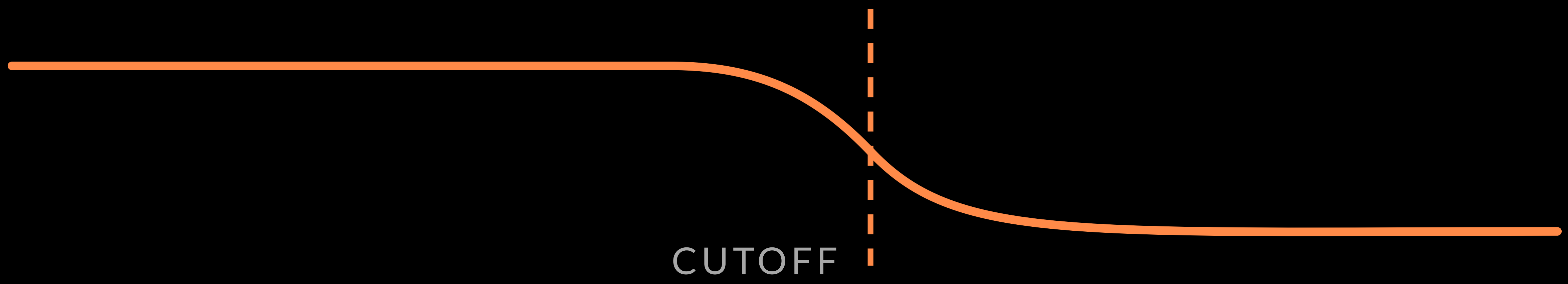
SYNC
Locks Oscillator A's cycle to Oscillator B for aggressive timbres.

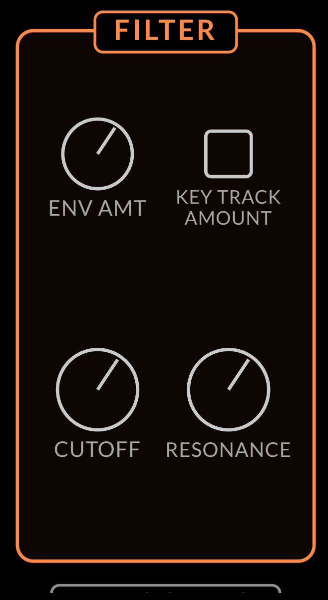
LO1 / LO2
Osc B's low modes: LFO-range pitch, key-tracked (LO1) or free (LO2).

HARMONICS
Saw has them all; square and triangle only the odd ones — the triangle's fade much faster.

FILTER

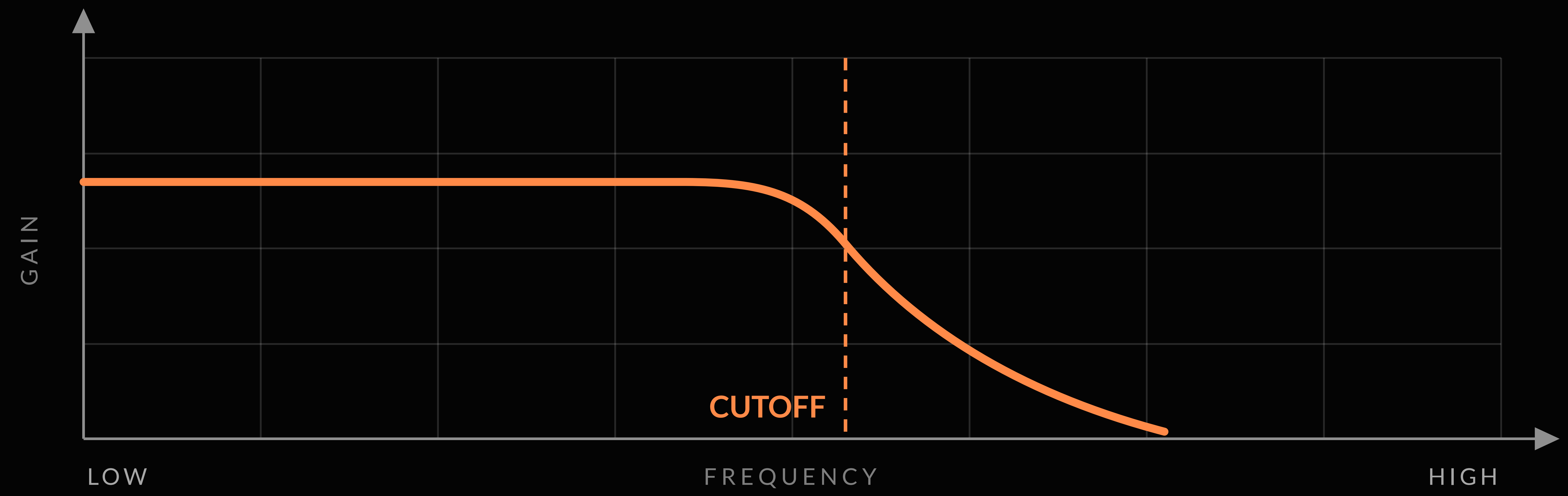
Shaping your tone





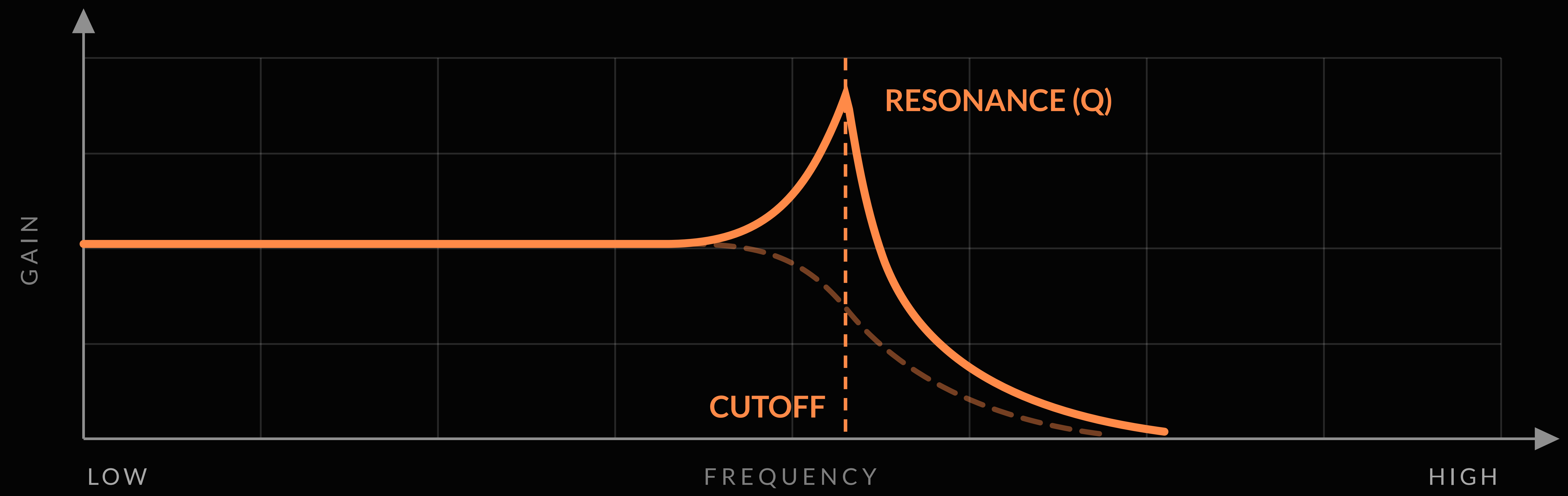
LOW-PASS

Below the cutoff passes, above it fades. Close it: darker. Open it: brighter.



RESONANCE

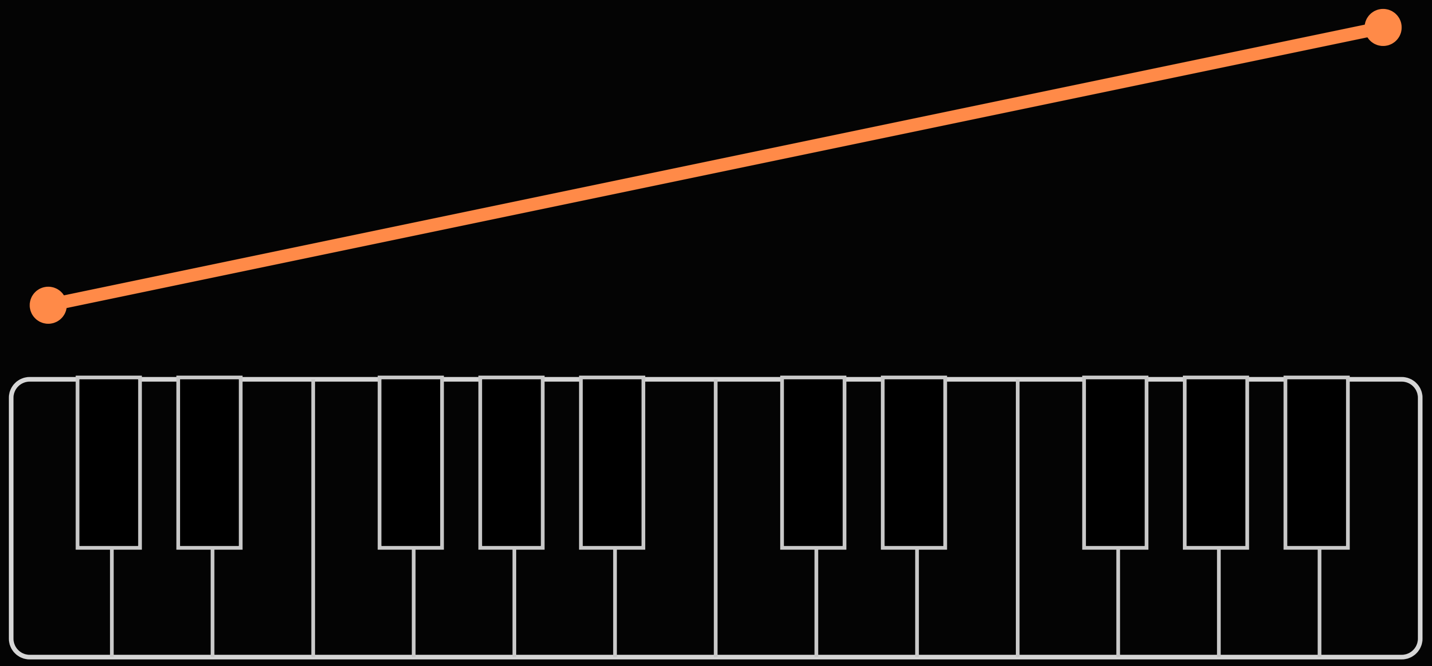
A boost right at the cutoff. Turn it up and the filter sings; at maximum it becomes a sound source of its own.



KEY TRACK & ENV AMOUNT

The keyboard opens the filter; the envelope moves it in time.

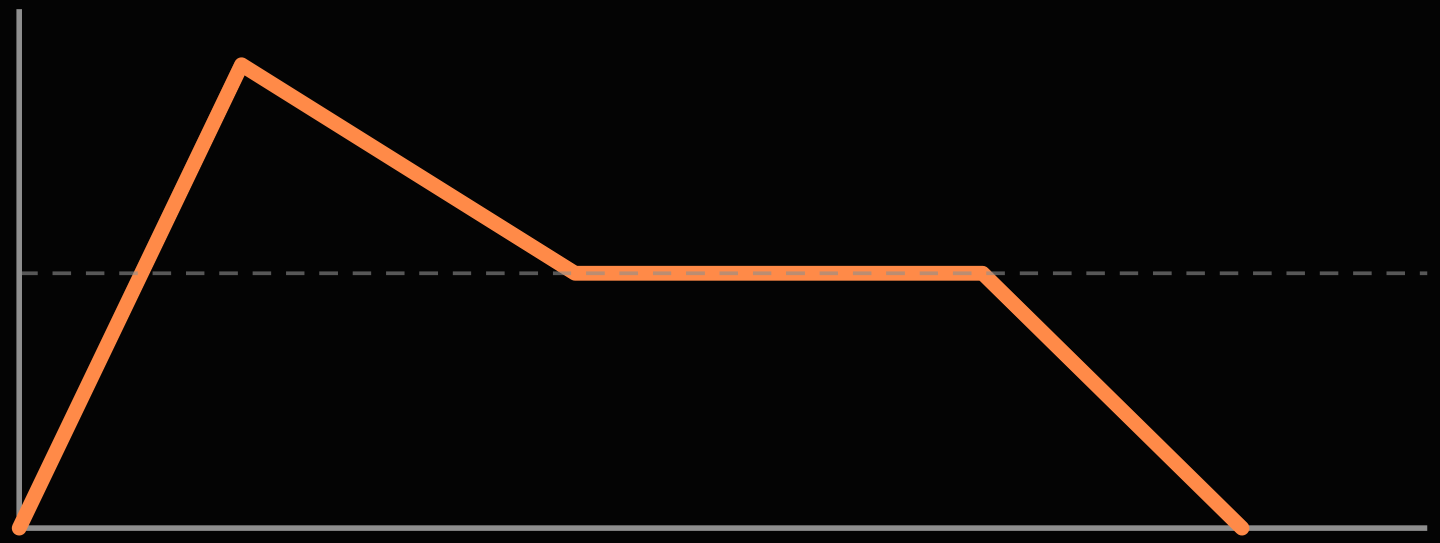
KEY TRACK



HIGHER NOTES, BRIGHTER FILTER

OFF · QUARTER · HALF · FULL

ENV AMT



THE FILTER ENVELOPE MOVES THE CUTOFF

OVER TIME · MODULE 04

KEY TERMS

New words from this module

CUTOFF
The hinge: below passes, above fades.

SELF-OSCILLATION
At maximum, the filter becomes a sine source.

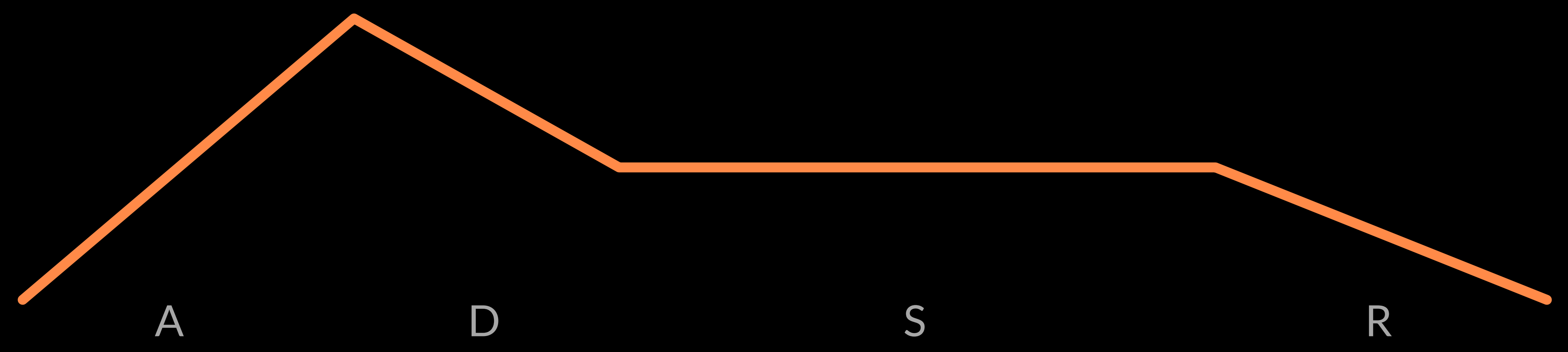
ENV AMOUNT
How much the envelope moves the cutoff.

RESONANCE
A boost right at the cutoff.

KEY TRACK
Higher notes, brighter filter.

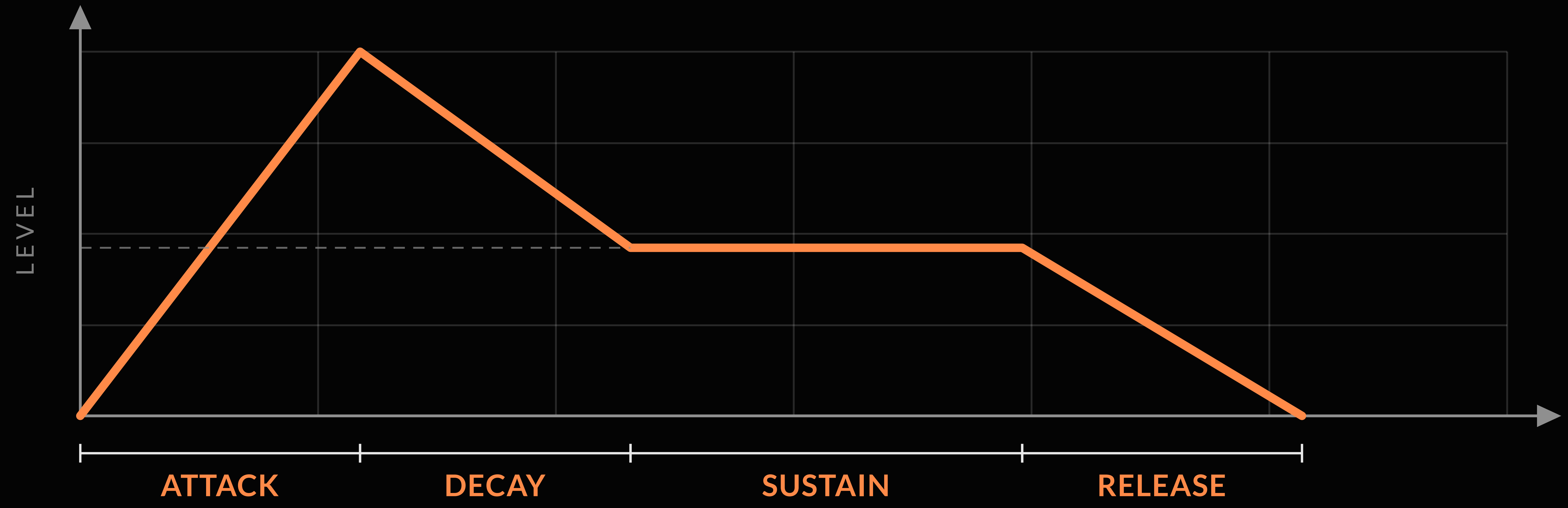
ENVELOPES

Controlling your sound over time



ADSR

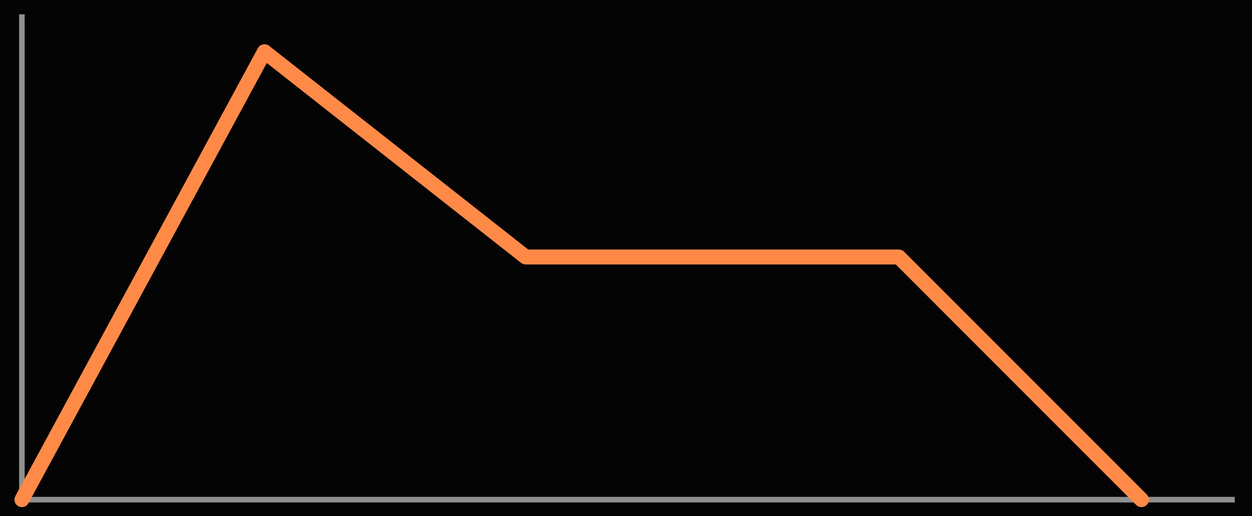
A shape that happens every time you press a key. Sustain is a level, not a time.



TWO ENVELOPES

Same shape, two jobs: how loud, how bright.

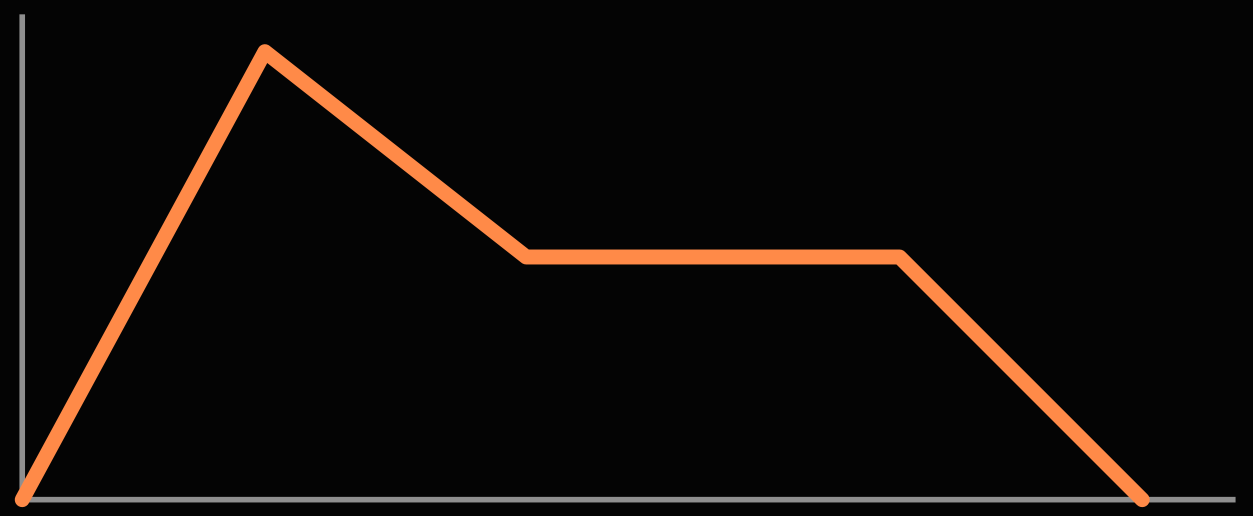
AMP ENVELOPE



LOUDNESS

HOW LOUD

FILTER ENVELOPE



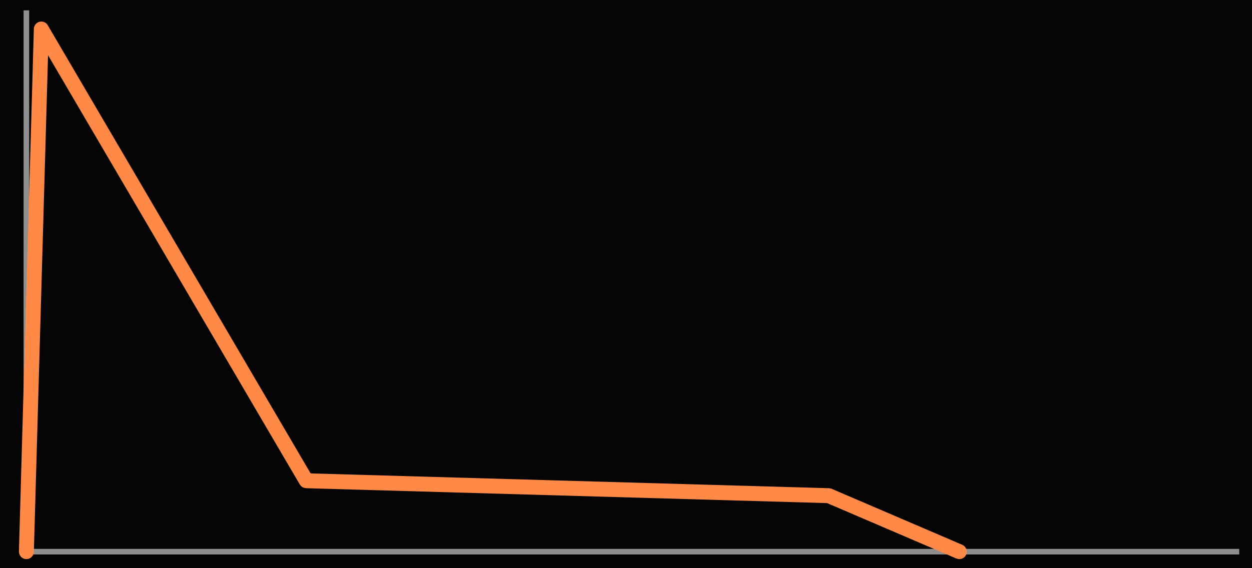
CUTOFF · SCALED BY ENV AMT

HOW BRIGHT

BASS vs PAD

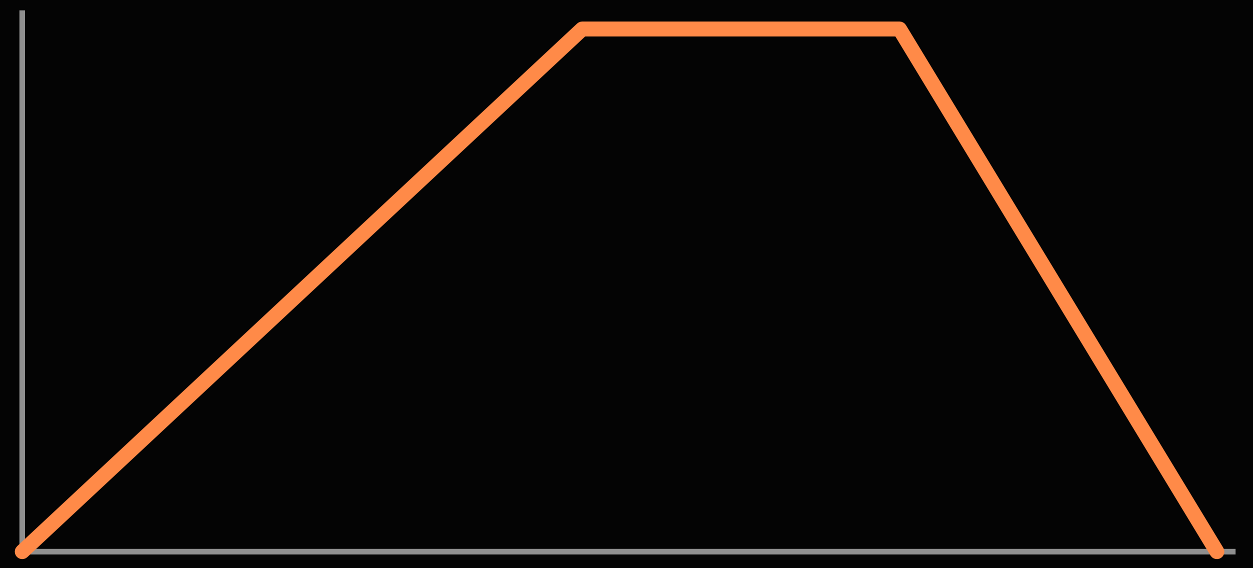
The same oscillators, two instruments.

BASS



INSTANT ATTACK · SHORT DECAY
LOW SUSTAIN

PAD



SLOW ATTACK · NO DECAY
FULL SUSTAIN · LONG RELEASE

KEY TERMS

New words from this module

ATTACK
Time from silence to the peak.

DECAY
Time from the peak to the sustain level.

SUSTAIN
A level, not a time.

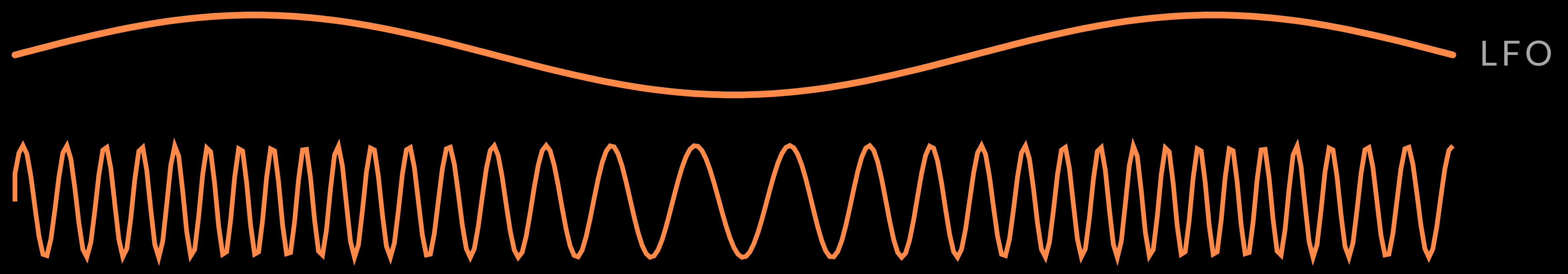
RELEASE
Time to fade after the key is let go.

ENV AMOUNT
How much the envelope moves the cutoff.

VELOCITY
How hard you strike scales the envelope.

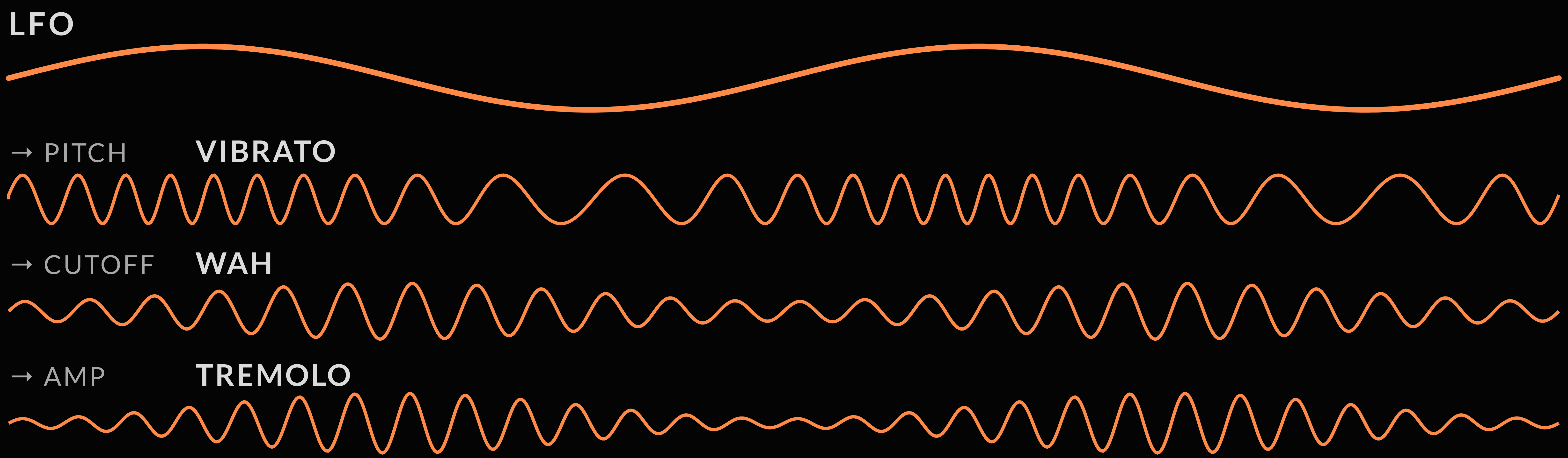
LFO & MODULATION

Adding movement



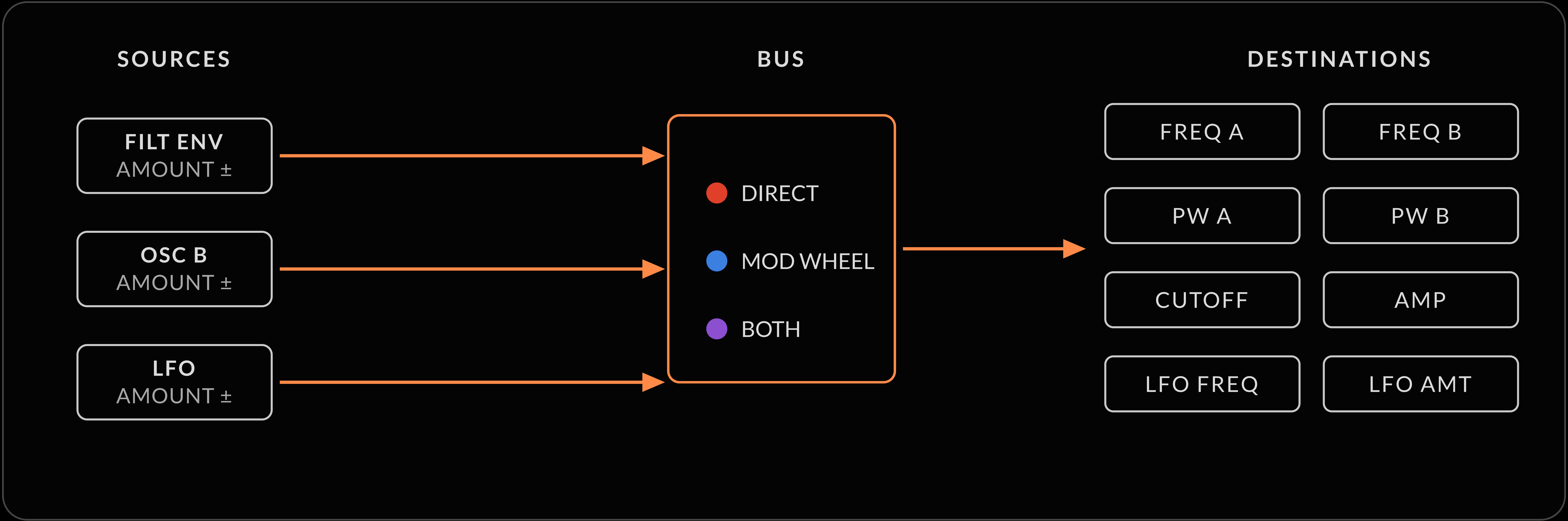
THE LFO

An oscillator too slow to hear. You hear what it moves.



THE MODULATION SECTION

Route → destination → amount. On the wheel bus, the mod wheel is the volume of the modulation.



KEY TERMS

New words from this module

LFO
An oscillator too slow to hear.

VIBRATO
LFO to pitch.

PWM
LFO to pulse width.

MOD WHEEL
The volume of the modulation.

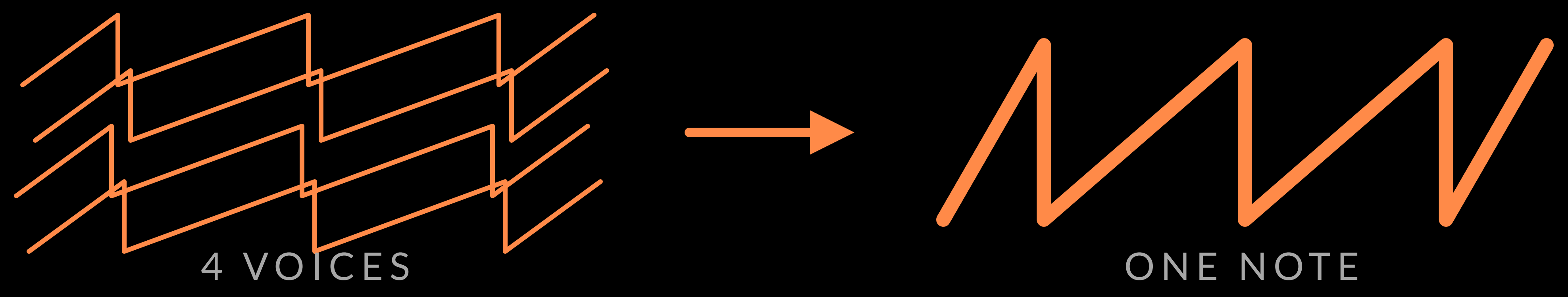
RATE
Its speed; SYNC locks it to the clock.

TREMOLO
LFO to amplitude.

MODULATION BUS
Direct (red) or mod wheel (blue).

VOICES & UNISON

Four voices, one note, and the loop that thickens everything

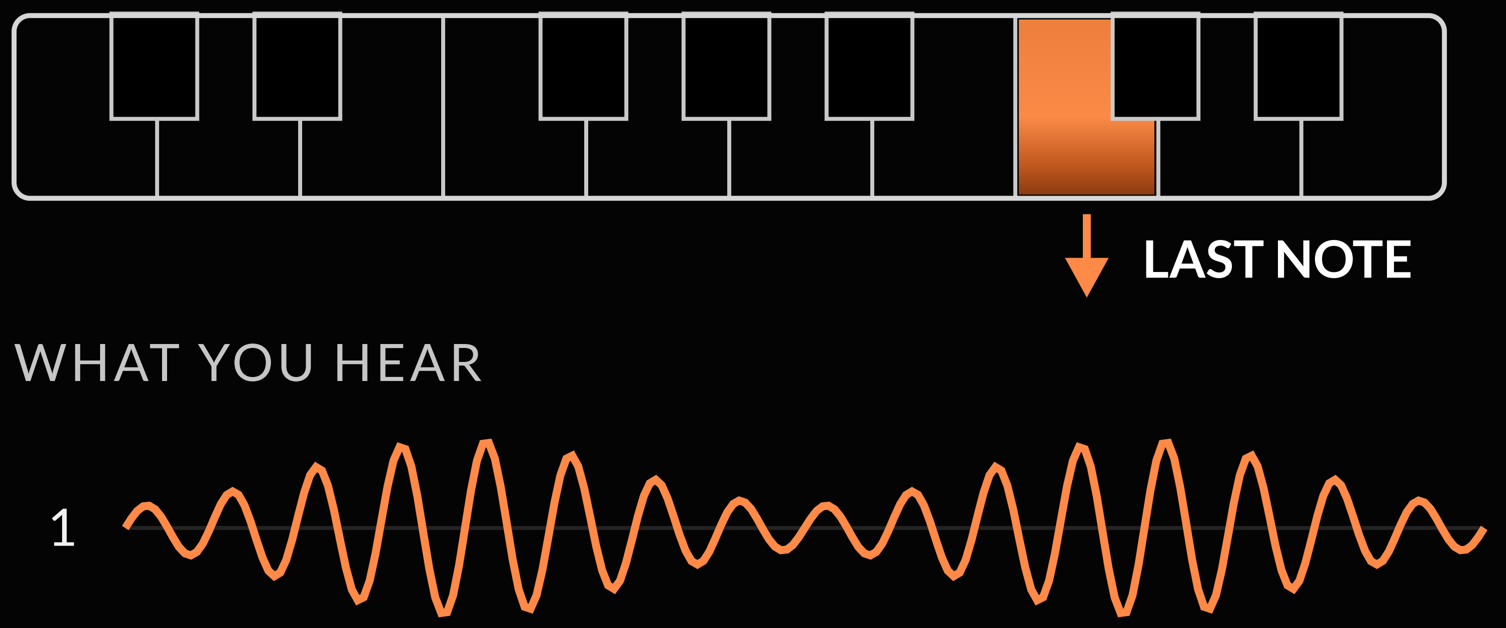


POLYPHONY

Fourm has four complete voices. Play a fifth note and the oldest one is stolen.

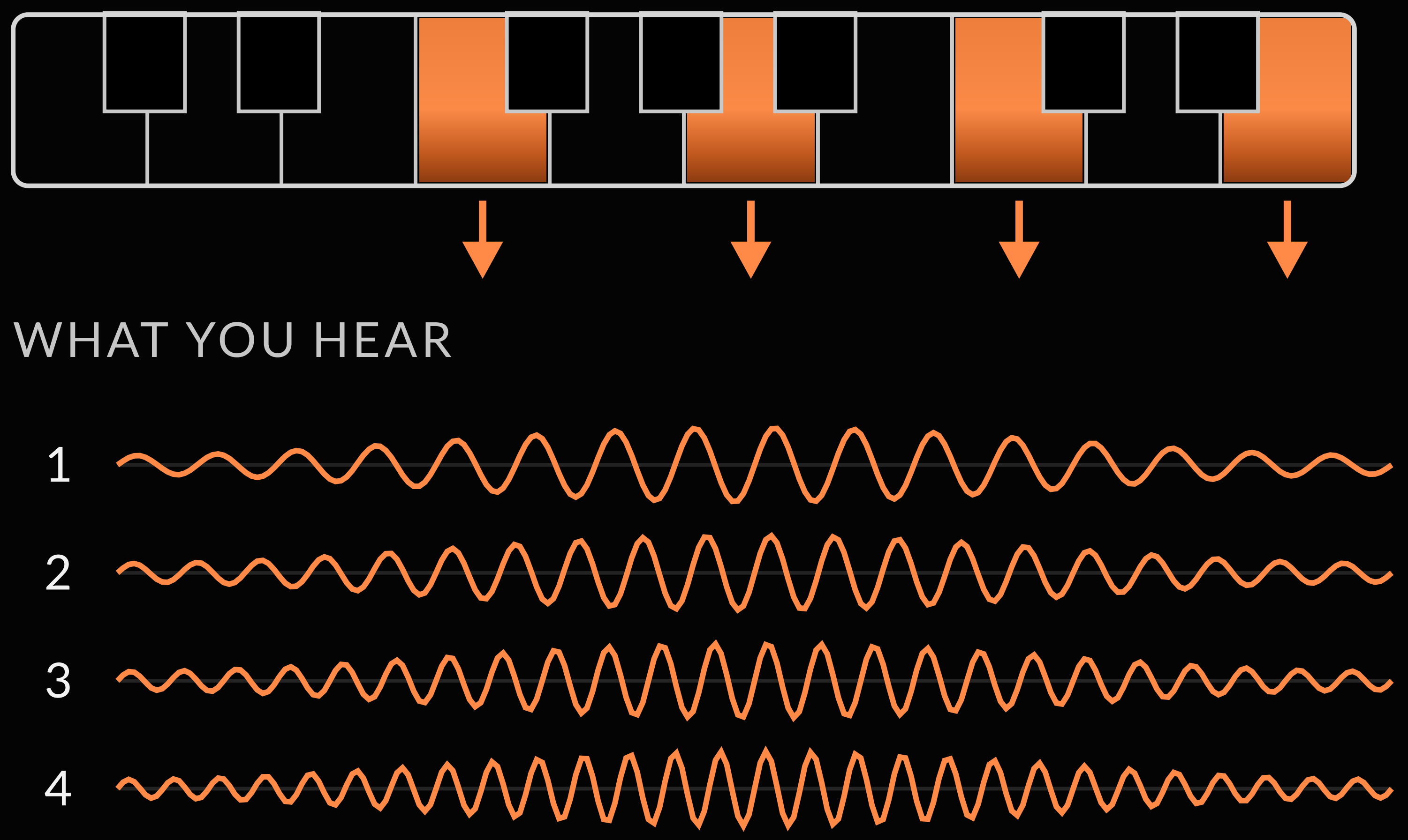
MONOPHONIC · ONE VOICE

WHAT YOU PLAY



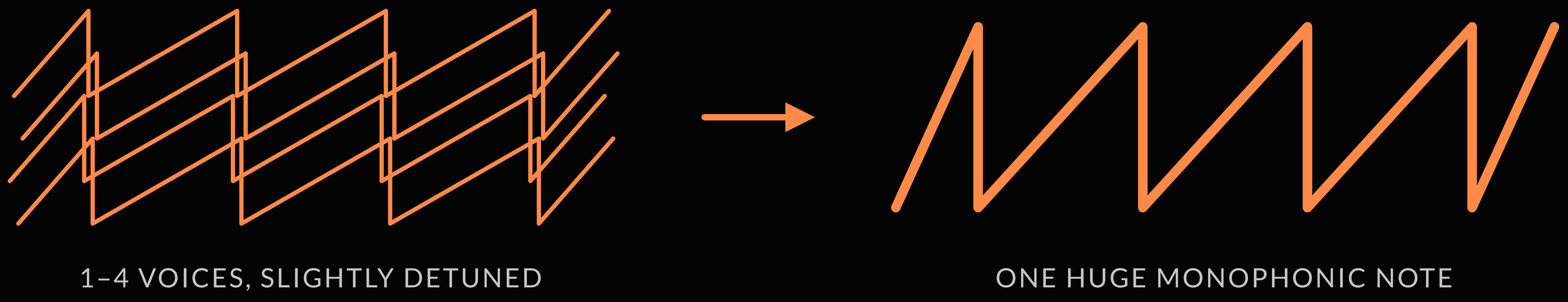
4-VOICE POLYPHONIC

WHAT YOU PLAY



UNISON

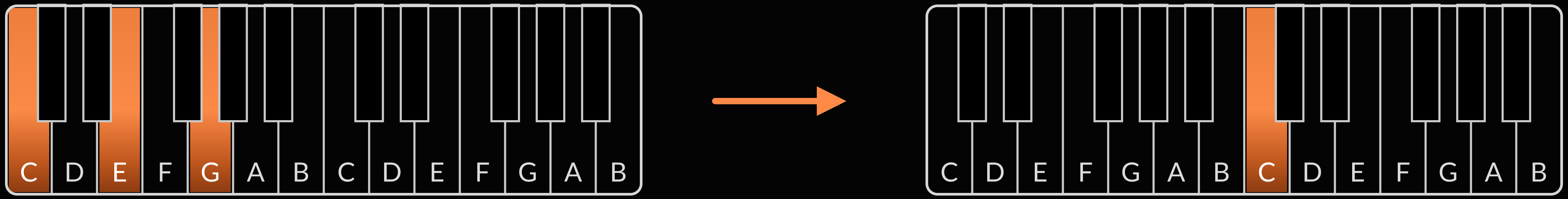
Four voices on one note



HOLD UNISON + SELECT FOR VOICES · PARAM 10: UNISON DETUNE 0–127

CHORD MEMORY

One key plays the whole chord



1 · HOLD A CHORD, PRESS UNISON

2 · ANY SINGLE KEY REPLAYS IT, TRANSPOSED

SAVED WITH THE PROGRAM: GREAT FOR CHORD STABS

FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

DEFAULT LEVEL 100 · HOLD FEEDBACK + TURN SELECT · THICKEN, DRIVE, CHAOS

KEY TERMS

New words from this module

UNISON
All chosen voices play the same note. Monophonic, massive.

CHORD MEMORY
Unison stores a chord; single keys replay it transposed.

FEEDBACK
The output fed back into the mixer, soft-clipped.

POLYPHONY · VOICE
Four complete voices; the fifth note steals the oldest.

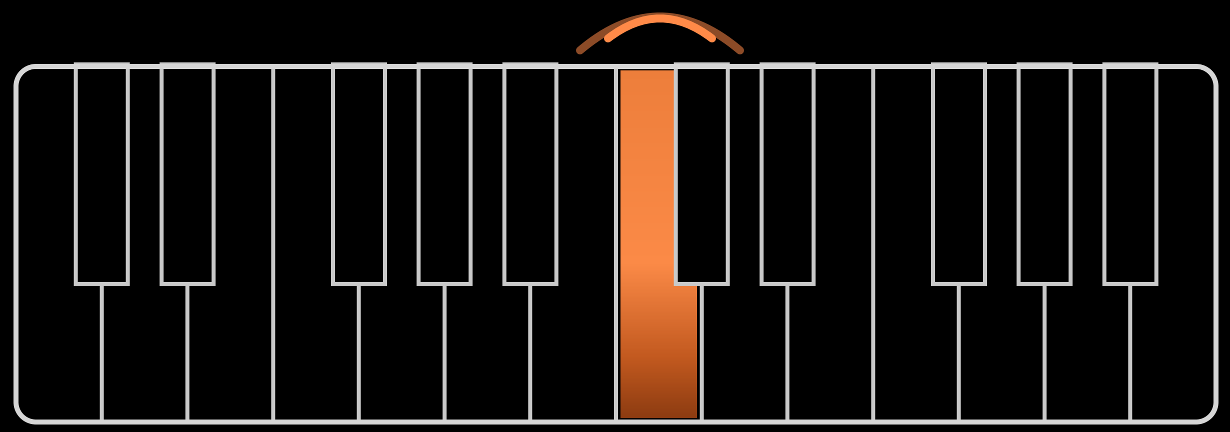
UNISON DETUNE
0–127 of drift between stacked voices: the chorused fat.

HOLD
Sustains up to four notes after your hands leave the keys; also latches the arpeggiator.

SOFT-CLIPPING
Gentle rounding of peaks: saturation instead of harsh distortion.

EXPRESSION

Play it, don't just press it

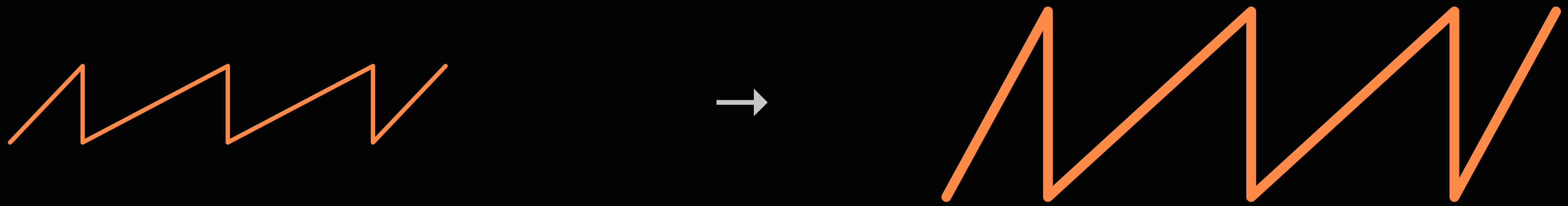


VELOCITY

How fast the key goes down. Scales the amp and filter envelopes; five response curves in Global.

SOFT

HARD



DARKER, QUIETER

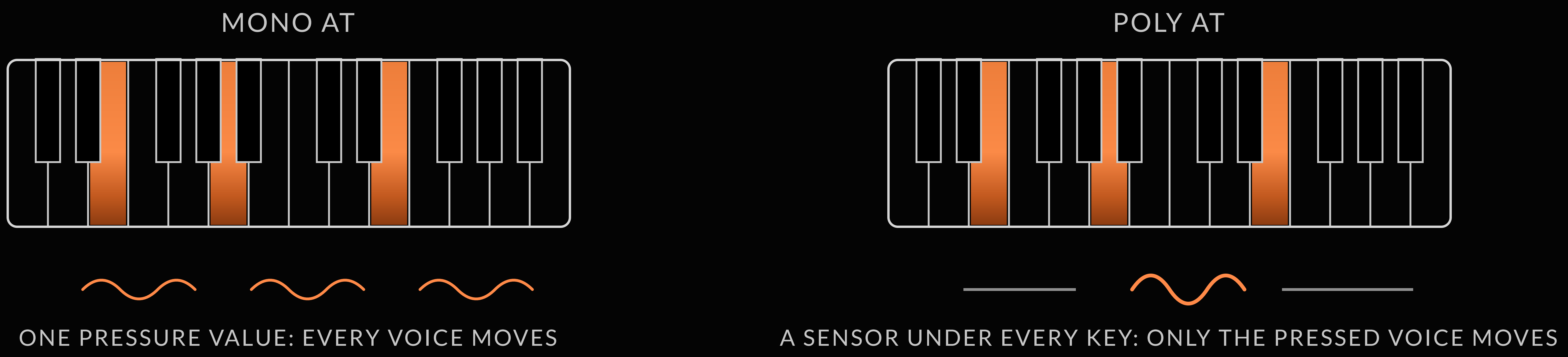
BRIGHTER, LOUDER

VELOCITY FEEDS BOTH ENVELOPES: FILT ENV VEL & AMP ENV VEL, 0–127

FIVE RESPONSE CURVES IN GLOBAL: SOFT+, SOFT, MEDIUM, HARD, HARD+

POLY AFTERTOUCH

Pressure, per key: the Tactive keybed



PRESSURE ROUTING

Press into one note of a chord and only that note opens.

FREQ A | FREQ B | FILT | AMP | LFO FREQ | LFO AMT

PICK DESTINATIONS WITH THE DEST BUTTONS: TWO PER BUS

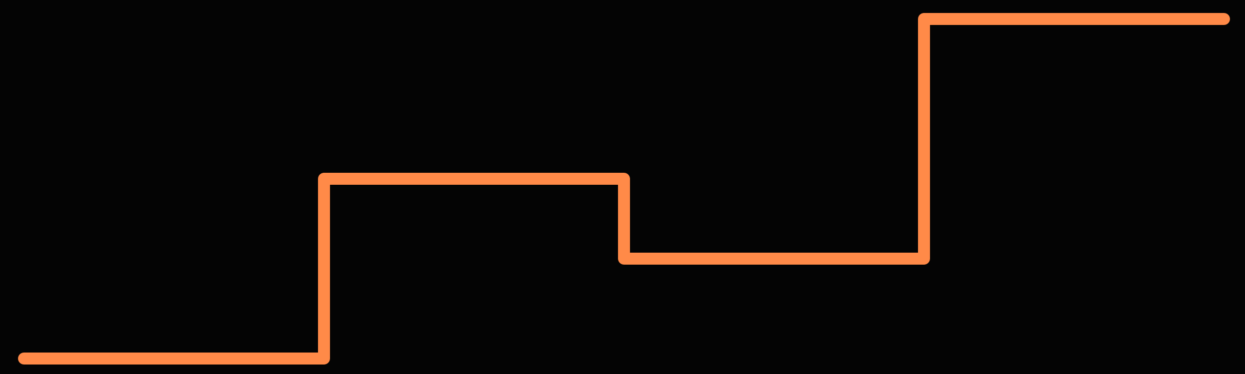
ONE AMOUNT KNOB, POSITIVE OR NEGATIVE

PRESS HARDER WITH +FILT: THE SOUND OPENS · WITH −FILT: IT CLOSES

GLIDE

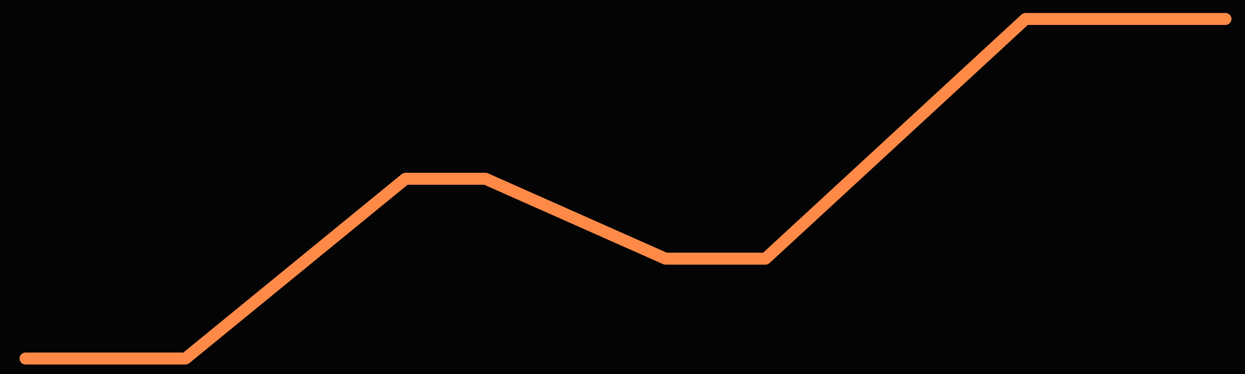
GLIDE on and RATE above zero. Modes: hold GLIDE and turn SELECT.

GLIDE OFF



PITCH JUMPS INSTANTLY

GLIDE ON + RATE



PITCH SLIDES BETWEEN NOTES

GLIDE ON WITH RATE AT 0 DOES NOTHING. SET BOTH

KEY TERMS

New words from this module

VELOCITY
How fast a key goes down. Routed to filter and amp envelope amounts.

POLY vs MONO AT
Per-key pressure vs one value for the whole keyboard.

CURVE
Five velocity curves, from Soft+ to Hard+; aftertouch has its own five.

GLIDE · PORTAMENTO
Pitch slides between notes instead of jumping; legato modes slide only on overlap.

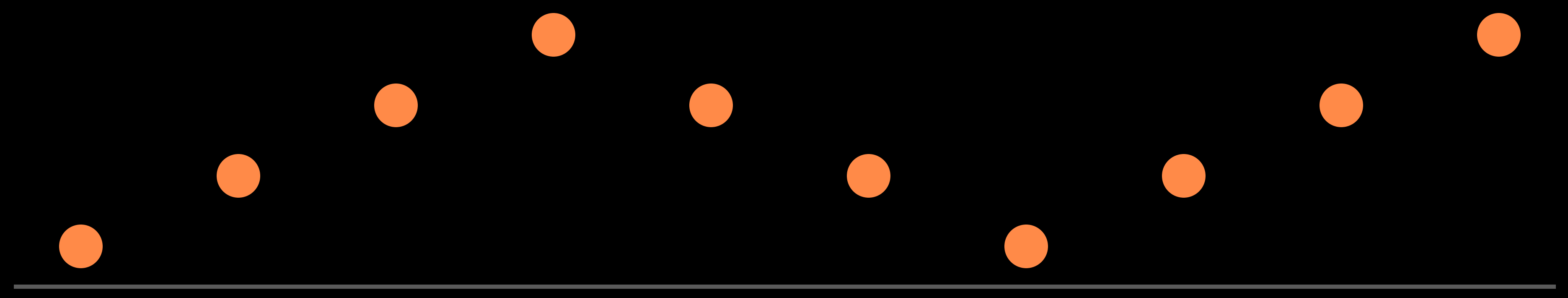
AFTERTOUCH (AT)
Pressure applied after the key bottoms out.

TACTIVE KEYBED
This synth's keybed: a silicone-enclosed pressure sensor per key.

AMOUNT ±
Modulation depth; negative amounts invert the direction.

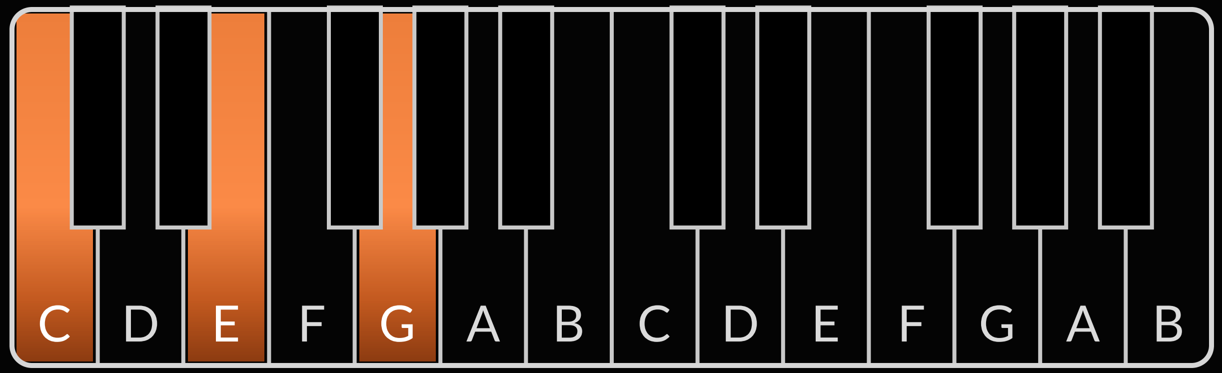
ARPEGGIATOR & SEQUENCER

Patterns from held notes

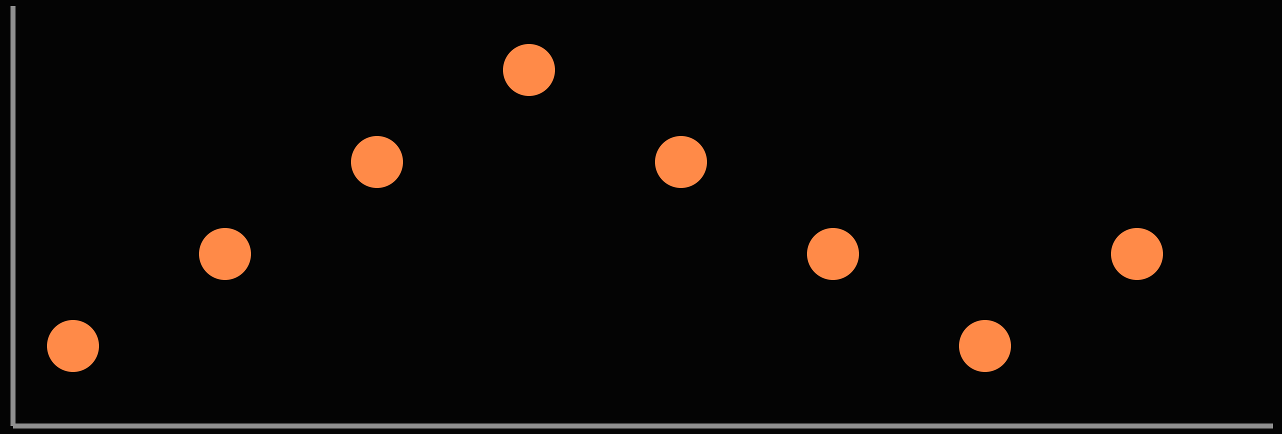


HOLD A CHORD

Hold a chord and the synth plays the pattern.



HOLD C · E · G



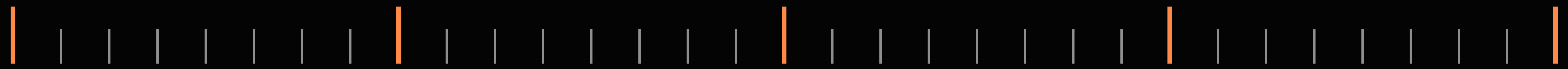
UP+DOWN 1 · OCTAVE 1

MODES: UP · DOWN · UP+DOWN 1 & 2 · RANDOM · ASSIGN. PRESS HOLD TO LATCH

CLOCK & SYNC

Tap TAP TEMPO four times, or hold it and turn SELECT. CLK DIV picks the grid.

Clock Mode set to In: the arp waits for external clock.



SET TEMPO: TAP 4× ON TAP TEMPO · OR HOLD BPM + SELECT

CLK DIV

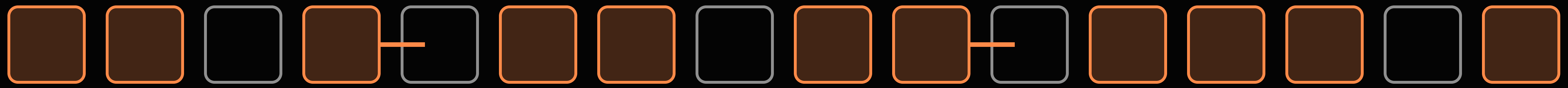
LFO SYNC

MIDI IN/THRU

WITH MIDI CLOCK MODE IN, THE ARP WAITS FOR EXTERNAL CLOCK

NOTE & MOD SEQ

A sequencer in disguise



RECORD UP TO 64 STEPS: NOTES, RESTS, TIES, EVEN GLIDES

SEQ MOD SENDS STEPS TO: OSC FREQ, CUTOFF, PW, FEEDBACK, MOD AMOUNTS

PLAY ANY KEY: THE WHOLE SEQUENCE TRANSPOSES

KEY TERMS

New words from this module

ARPEGGIATOR
Plays a pattern from the notes of a held chord.

RELATCH
Each new chord starts a fresh arpeggio instead of adding notes.

CLOCK DIVISION
Note value per step: quarters, eighths, sixteenths…

MOD SEQUENCE
Steps that move a parameter instead of playing notes.

HOLD / LATCH
Keeps the arpeggio running after you release the keys.

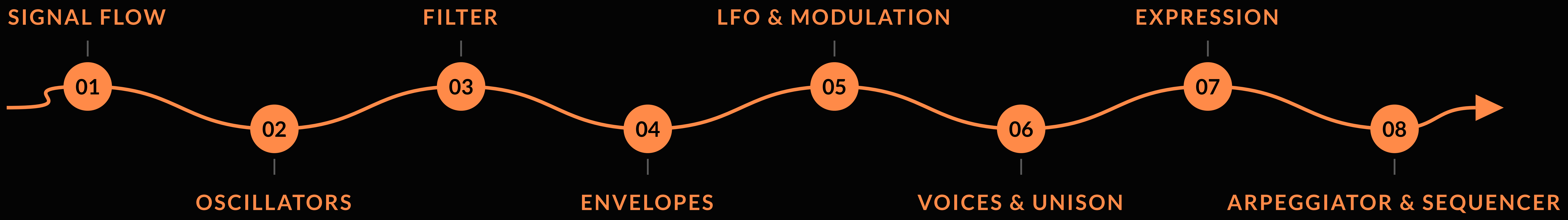
BPM / TAP TEMPO
The clock speed: dialed in or tapped in.

REST · TIE
A silent step · a step that extends the previous note.

SEQ REVIEW
Global setting for stepping through and editing a sequence.

THE ROAD, AGAIN

Every synth is this road with different clothes.



THANK YOU

Take the handbook home — EN or RO

SINTEZAUR

community & next workshop

QR

ZEEDO

try and buy the Fourm

QR

SEQUENTIAL

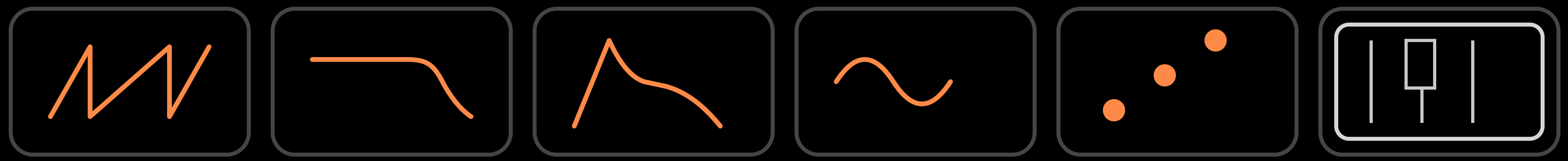
user guide & presets

QR

SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

GLOSSARY

The full glossary is in your handbook



← COURSE MAP